Answer key: Lesson 5, Equal Groups

For the tutor and grown-ups. Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

Start here (pages 1 and 2)

S1. Are the groups equal? Circle Yes or No for each picture.

Answer: Left: Yes. Right: No.

The left picture has 3 bags with 4 sweets in each bag: the groups are equal. The right picture has 2, 3 and 2 cookies: the groups are not equal.

Watch for: Counting only the number of groups.; Saying Yes because the pictures look alike.

S2. Look at the picture. Fill in the blanks.

Answer: 3 plates; 5 cookies on each plate; 3 groups of 5 = 15

There are 3 plates and each plate has 5 cookies. 5 + 5 + 5 = 15, so 3 groups of 5 = 15.

Watch for: Writing 5 groups of 3.; Counting the cookies on every plate and losing count.

S3. Mei has 5 jars. Each jar has 2 sweets. How many sweets does Mei have in all?

Answer: 2 + 2 + 2 + 2 + 2 = 10; 5 groups of 2 = 10

5 jars with 2 sweets each: 2 + 2 + 2 + 2 + 2 = 10. So 5 groups of 2 = 10 and Mei has 10 sweets in all.

Watch for: Adding 5 + 2 = 7.; Losing count of the jars.

S4. Draw 4 dots on each plate. Then fill in the blank.

Answer: 3 groups of 4 = 12

3 plates with 4 dots on each: 4 + 4 + 4 = 12, so 3 groups of 4 = 12.

Watch for: Drawing 3 dots on 4 plates.; Drawing different numbers of dots on the plates.

S5. Fill in the table.

Answer: 15, 8, 10

3 groups of 5 = 5 + 5 + 5 = 15. 4 groups of 2 = 2 + 2 + 2 + 2 = 8. 2 groups of 5 = 5 + 5 = 10.

Watch for: Adding the two numbers: 3 + 5 = 8.; Writing the number of groups in the Total box.

S6. Aisha has 4 boxes. Each box has 5 pencils. How many pencils does Aisha have in all?

Answer: 4 × 5 = 20 (5 × 4 = 20 is also right); 20 pencils

4 boxes with 5 pencils in each box. 5 + 5 + 5 + 5 = 20. The number sentence is 4 × 5 = 20 and Aisha has 20 pencils.

Watch for: Writing + in the circle.; Writing 4 × 5 = 9.

Think harder (page 3)

T1. Look at the dots in rows. Fill in the blanks.

Answer: 2 × 8 = 16; 8 × 2 = 16; Yes. (In (a) 8 × 2 = 16 is also right if the child counted the dots in one row first; then (b) is 2 × 8.)

The picture has 2 rows with 8 dots in each row: 2 × 8 = 16. Turned on its side it has 8 rows with 2 dots in each row: 8 × 2 = 16. The totals are the same, so 2 × 8 = 8 × 2.

Watch for: Adding 2 + 8 = 10.; Thinking the totals must be different because the pictures look different.

T2. Each plate should have 3 cookies. Look at the picture. How many more cookies are needed to make the groups equal?

Answer: 3 more cookies

The plates have 3, 1 and 2 cookies. The second plate needs 2 more and the third plate needs 1 more. 2 + 1 = 3 more cookies.

Watch for: Writing 6 (the number of cookies now).; Adding only the missing cookies of one plate.

T3. Kumar has 3 plates with 4 apples on each plate. He says he has 7 apples.

Answer: 3 groups of 4 = 12; No

3 groups of 4 apples: 4 + 4 + 4 = 12. 7 is 3 + 4, which is not the same as 3 groups of 4. So Kumar is not correct.

Watch for: Saying Yes because 3 + 4 = 7.; Writing 3 × 4 = 34.

T4. Jenny has 12 stickers. She puts them into equal groups in different ways. Fill in the table.

Answer: 3, 3

4 groups of 3 = 12, so each of the 4 groups has 3. 3 groups of 4 = 12, so the number of groups is 3.

Watch for: Writing the total in every box.; Choosing numbers that do not make 12.

Go further (page 4)

G1. A shed has 3 bikes and 4 cars. Each bike has 2 wheels. Each car has 4 wheels. How many wheels are there in all?

Answer: 6 wheels, 16 wheels, 22 wheels

3 bikes have 2 + 2 + 2 = 6 wheels. 4 cars have 4 + 4 + 4 + 4 = 16 wheels. 6 + 16 = 22 wheels in all.

Watch for: Counting a car as 2 wheels.; Forgetting to add the wheels of the bikes and the cars.

G2. Jamie has 6 jars. Each jar has 5 marbles. Jamie needs 40 marbles. How many more marbles does he need?

Answer: 6 × 5 = 30; 10 more marbles

6 jars with 5 marbles in each jar: 5 + 5 + 5 + 5 + 5 + 5 = 30 marbles. 40 - 30 = 10 more marbles.

Watch for: Writing 40 - 6.; Stopping after finding 30.

G3. Wei has 24 sweets. Show 3 different ways to put them into equal groups. Use a different number of groups each time. Many answers are right.

Answer: For example 2 groups of 12, 3 groups of 8, 4 groups of 6 (also 6 of 4, 8 of 3, 12 of 2, 1 of 24, 24 of 1)

2 × 12 = 24, 3 × 8 = 24, 4 × 6 = 24. Turning each one gives 12 × 2, 8 × 3 and 6 × 4.

Watch for: Choosing numbers that do not make 24.; Writing the same way twice.

Investigation (page 5): Rows with 12 counters

Six ways: 1 × 12, 2 × 6, 3 × 4, 4 × 3, 6 × 2 and 12 × 1. Turning a picture gives a second one (2 × 6 and 6 × 2). 5 rows does not work: 5 rows of 2 uses 10 counters and 2 are left over.

Look for: A child who builds several arrays, writes a × sentence for each and notices that turning one gives another. Credit a child who tries 5 rows and finds counters left over. Do not mark a child down for finding only three ways.

Exit ticket (page 5)

E1. Fill in the blank.

Answer: 20

4 fives: 5 + 5 + 5 + 5 = 20, so 4 groups of 5 = 20.

Watch for: Writing 9 (5 + 4).; Losing count of the fives.

E2. Which is the same as 4 groups of 3? Write its number in the bracket.

Answer: (2) 3 + 3 + 3 + 3

4 groups of 3 means 3 four times: 3 + 3 + 3 + 3.

Watch for: Choosing 4 + 3.; Choosing 3 + 3 + 3.

E3. Ali has 3 boxes. Each box has 4 pencils. How many pencils does Ali have in all?

Answer: 3 × 4 = 12 (4 × 3 = 12 is also right); 12 pencils

3 boxes with 4 pencils in each box: 4 + 4 + 4 = 12. The sign in the circle is ×.

Watch for: Writing + in the circle.; Writing 3 + 4 = 7.

Sneak-peek puzzles (not in the workbook)

1. 6
2. Same. Each of them has 12 sweets: 4 groups of 3 and 3 groups of 4 are both 12.